

Danium S. Syed

Applied Cybersecurity

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Summary

Passionate Computer Engineer and Erasmus Mundus Scholar. Solid experience in DevOps, System Administration, and Artificial Intelligence security research. Currently completing a double Master's degree in Cybersecurity and holding a CCNA certification.

Education

Kadir Has University & SRH Berlin University of Applied Sciences, Erasmus Mundus Double Master's in Applied Cybersecurity (CyberMACS) – Istanbul & Berlin Sept 2024 – Sept 2026

- Selected among the top 32 students from over 1,000 applications
- Received a fully funded scholarship given to the top applicants
- GPA: 3.48/4.00 (KHAS), 1.8 (SRH)

American International University-Bangladesh (AIUB), BSc in Electrical and Electronics Engineering – Dhaka, Bangladesh Sept 2019 – Sept 2023

- Received multiple awards and studied on a full scholarship since admission
- Graduated top of class with a CGPA of 3.98/4.00 (1.03 German scale)
- Received Best Project Award for Thesis/Capstone Project

Mount Royal Academy, Pearson Edexcel IGCSE and A-Levels in International British Curriculum – Bangladesh June 2017 – June 2019

Experience

Cybersecurity Intern, Cyberforce – Istanbul, Turkey June 2025 – Sept 2025

- Developed internal tools for security testing of LLM/ML models from PyTorch, TensorFlow, Keras, Sklearn, and XGBoost, detecting insecure serialization formats inside Docker containers and malware-corrupted models to prevent model serialization attacks
- Conducted on-site and offline penetration testing of ML and LLM models; completed all levels of the Gandalf and Prompt Airlines CTFs
- Gained practical knowledge of ISO 27001 through internship training
- Penetration tested various HackTheBox systems, gaining practical knowledge of web security

System Administrator, SECOM Technologies Ltd. – Dhaka, Bangladesh Mar 2023 – Aug 2024

- Managed Linux, macOS, and Windows virtual machines supporting internal business applications
- Administered Proxmox virtualization infrastructure and used Ansible playbooks

- ICT and DevOps Intern**, Grameenphone (Telenor) – Dhaka, Bangladesh Jan 2023 – June 2023
- Automated ICT workflows and built an internal tool to monitor monthly product revenues across 8 products, presenting them on a dashboard
 - Worked on the system architecture formation of Grameenphone Alo, a proprietary IoT platform
 - Extensively worked on hybrid-cloud architectures and Kubernetes, and on secure CI/CD deployment through Jenkins, Docker, and Terraform
 - Conducted on-site surveys of Grameenphone datacenters

- IT Educator**, Dignitus International School – Narayanganj, Bangladesh July 2017 – July 2019
- Educated students from grade 8 to 10, with multiple A* achievements in taught subjects for board exams
 - Maintained lesson plans and liaised with the international education board Edexcel

Projects

- Master's Thesis: Security Analysis of Machine Learning Artifacts in Containerized AI Supply Chains** Jan 2026 – present
- Conducted security analysis of containerized ML systems to measure prevalence and distribution of embedded model artifacts across Docker ecosystems
 - Identified security risks in ML supply chains by statically analyzing model files (PyTorch, TensorFlow, ONNX, Pickle) for unsafe serialization and code execution vulnerabilities
 - Built an automated pipeline to extract container filesystems and scan ML artifacts, generating empirical insights into real-world deployment risks

- Linux Home Server (HomeLab)** Ongoing
- Built from scratch and maintains a Debian-based home server for security research, home automation, and media consumption
 - Follows DevOps practices and maintains high uptime
 - Technology stack: WireGuard, Docker Compose, Proxmox

- Bachelor's Capstone: Long-Range Off-Grid Communication System Deployed with Amphibious Rovers** Jan 2023 – Sept 2023
- Built a mesh communication system using LoRa modulation for deployment in disaster scenarios via amphibious rovers
 - Provides text messaging and location-based communication services for affected users in disaster zones

Publications

- Design and Feasibility Analysis of a LoRa Based Communication System for Disaster Management** Aug 2024
Danium S. Syed, S. B. Hossain, A. Islam, T. A. I. Jim, M. Saniat, R. Zishan
[10.53799/ajse.v23i2.1055](#) (AIUB Journal of Science and Engineering (AJSE), 23, 135-144)
- Reputation and Energy-Aware Dynamic Hybrid Consensus (READ-HC) Model for IIoT** July 2025
S. Mitra, V. S. Sable, Danium S. Syed
[10.31585/jbba-8-2-\(3\)2025](#) (The Journal of The British Blockchain Association)
- Explainable Botnet Detection Using Flow-Based Random Forest Classification and OSINT Validation** Apr 2026
A. Razan, E. O. Njoku, Danium S. Syed, N. Nachevski
(Proc. CIIT 2026)

Analyzing Political Sentiment on Twitter with Zero-Shot Large Language Models

Oct 2026

R. Ayoub, *Danium S. Syed*, E. O. Njoku
(IACyC 2026)

Security Analysis of Machine Learning Artifacts in Containerized AI Supply Chains

Nov 2026

Danium S. Syed

(Submitted to Springer Lecture Notes in Computer Science (LNCS) conference proceedings)

Honors and Awards

- 5th Place (All Solves), CYBERUS Spring School CTF Competition — Lorient, France (2026)
- 2nd Place (All Solves), CyberMACS Winter School CTF Competition — Istanbul, Turkey (2025)
- Erasmus Mundus Full Scholarship, EMJM in Applied Cybersecurity — Turkey/Germany (2024)
- Dean's Best Project Award, AIUB — Dhaka, Bangladesh (2023)
- Dean's List Scholarship, AIUB — Dhaka, Bangladesh (2019)
- Freedom Fighter's Scholarship, AIUB — Dhaka, Bangladesh (2019)
- Edexcel High Achiever's Award, IGCSE/Grade 10 — Dhaka, Bangladesh (2017)
- Featured in The Daily Star and The Daily Sun, IGCSE/Grade 10 — Dhaka, Bangladesh (2017)

Certifications

- Building Resilience: Navigating Vulnerability, Risks, and Continuity in a Dynamic World — CyberMACS (2026)
- Cisco Certified Network Associate (CCNA), CSC014934967 — Cisco (2025)
- The Bits and Bytes of Computer Networking — Google/Cisco, Coursera (2020)

Skills

DevOps: AWS, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions

System Administration: Linux, Networking, Virtualization, Containers, Git

Privacy & AI Security: Differential Privacy, Agentic AI, PyTorch, TensorFlow, vLLM/Ollama

Security Analyst: Microsoft Sentinel, IDS, Threat Intelligence (MITRE ATT&CK), Incident Response, Jira

Scripting: Python, JavaScript, Bash, LaTeX

Languages: English (IELTS 8.0), Bengali (Native), Turkish (A1), German (A1, ongoing studies)